

The Nobel Prize in Physiology or Medicine is given for discoveries deemed particularly important in the fields of physiology or medicine. In 1906, this award was given to two biologists for their groundbreaking work in the field of neuroanatomy. Despite using the same technique to visualize neurons, the two biologists formed opposing theories regarding organization of the nervous system.

Hypothesis 1: Reticular Theory

Biologist 1 developed a neural staining technique that was carried out by fixing brain tissue from older animals with primarily myelinated nerve fibers in potassium dichromate ($K_2Cr_2O_7$) before treating this tissue with silver nitrate ($AgNO_3$). A silver chromate (Ag_2CrO_4) precipitate deposited within the neurons, making them appear black. This silver impregnation method was revolutionary because of two features. First, only a random subset of neurons within a given tissue sample (~1% of the total cell population) was subject to silver impregnation. Second, the impregnated cells were stained in their entirety, allowing for visualization of the complete neuron. Biologist 1 concluded that his findings supported the idea that the nervous system is a single reticulum of fused neural processes that transmit information via cytoplasmic continuity.

Hypothesis 2: Neuron Doctrine

Biologist 2 modified the silver impregnation method by first subjecting the brain tissue to two rounds of staining ("double impregnation") rather than one. Where Biologist 1 studied the brain tissue of older animals with myelinated nerve fibers, Biologist 2 employed the silver stain on brain tissue from younger animals, which made it easier for the silver to impregnate the neural processes due to a relative lack of myelination. Biologist 2 reasoned that the nervous system is not a single web of interconnected fibers but rather is made up of discrete neural units that communicate via close, but not continuous, contact.

The neuron doctrine was later proven correct with the advent of electron microscopy, which allowed scientists to visualize the synapse (ie, the point of communication between two distinct neurons). Further investigation allowed researchers to ultimately discover that there are two types of synapses: electrical and chemical.

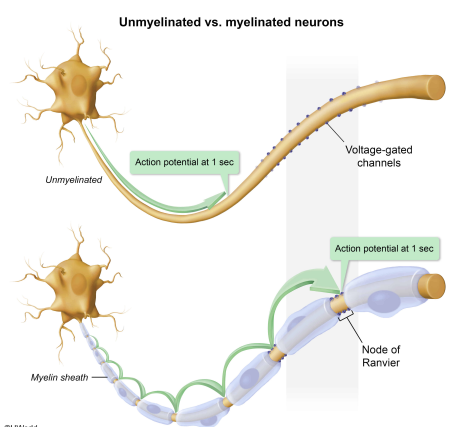
Compared to the axons investigated by Biologist 2, the axons studied by Biologist 1 most likely had:

- ☐ A. faster decay of the propagating electrical signal. [23%]
- ☐ B. axonal membranes that were more exposed to the extracellular space. [13%]
- ☐ C. slower rates of action potential conduction. [16%]
- ✓ ☒ D. voltage-gated ion channels clustered only at specific locations. [46%]

Correct

45% answered correctly

Explanation:



The **myelin sheath** is a multilayered, **segmented** covering that surrounds and insulates axons to **increase the conduction speed** of an action potential (AP). The sheath is produced by **glial cells** that wrap their plasma membranes tightly around the axon. Because myelin is a lipid substance, the sheath functions as an **insulator** (fat is a poor conductor of electric current) that keeps the current within the axon by preventing the dissipation of electric charge across the membrane.

However, the insulation is not perfect, and some charge does leak across the sheath back into extracellular space, weakening the current as it propagates down the axon. To ensure the membrane potential does not fall below the threshold before it reaches the axon terminal, the AP is regenerated at the nodes of Ranvier, which are gaps in the sheath where voltage-gated ion channels are concentrated. This type of conduction, where the **AP "leaps" from node to node** rather than flows down the entirety of the axon, is called **saltatory conduction**.

In unmyelinated axons, ions quickly dissipate back into the extracellular space, causing rapid weakening and decay of the propagating electrical signal. Accordingly, unmyelinated axons have voltage-gated ion channels positioned along the entirety of the axon to continually regenerate the AP. However, it takes time for each channel to open and for ions to move inward, so this repeated AP regeneration greatly decreases the speed of the signal. Other than myelin, increasing axon diameter can also increase AP speed by reducing intracellular electrical resistance.

According to the passage, **Biologist 1 studied** brain tissue with neurons containing primarily **myelinated axons** whereas Biologist 2 studied unmyelinated axons. Therefore, the *myelinated* axons investigated by Biologist 1 would **contain** voltage-gated **ion channels clustered at the nodes of Ranvier**. However, the unmyelinated axons investigated by Biologist 2 would contain voltage-gated ion channels that are more *evenly distributed* across the entire axon.

In addition, the myelinated (insulated) axons studied by Biologist 1 would exhibit slower (*not* faster) weakening of the electric signal (**Choice A**), have membranes with less (*not* more) exposure to the extracellular environment (**Choice B**), and would conduct APs at faster (*not* slower) rates (**Choice C**).

Educational objective:

The myelin sheath increases the speed of action potential (AP) propagation by acting as an electrical insulator that prevents dissipation of charge across the membrane. APs in myelinated axons travel via saltatory conduction.

Time spent:0 sec

Subject:
Biology

QID:400950

System:
Endocrine and Nervous Systems

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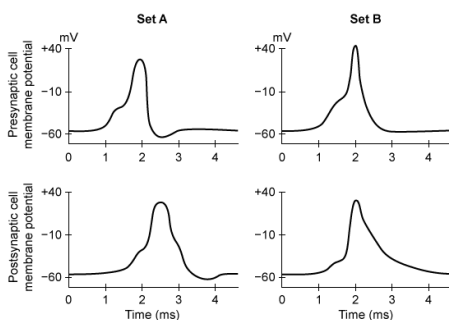
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Hypothesis 2: Neuron Doctrine

Biologist 2 modified the silver impregnation method by first subjecting the brain tissue to two rounds of staining ("double impregnation") rather than one. Where Biologist 1 studied the brain tissue of older animals with myelinated nerve fibers, Biologist 2 employed the silver stain on brain tissue from younger animals, which made it easier for the silver to impregnate the neural processes due to a relative lack of myelination. Biologist 2 reasoned that the nervous system is not a single web of interconnected fibers but rather is made up of discrete neural units that communicate via close, but not continuous, contact.

The neuron doctrine was later proven correct with the advent of electron microscopy, which allowed scientists to visualize the synapse (ie, the point of communication between two distinct neurons). Further investigation allowed researchers to ultimately discover that there are two types of synapses: electrical and chemical.

The graphs below depict neural signaling across two different kinds of synapses. One set corresponds to electric signal transmission, and the other corresponds to chemical signal transmission.



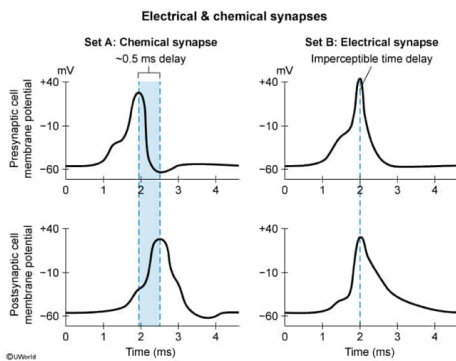
Based on this information, does the data in Set A depict action potential transmission across a chemical synapse or an electrical synapse?

- ✓ ☒ A. Chemical, because molecular signals must diffuse across the intercellular space [60%]
- ☐ B. Chemical, because one presynaptic potential leads to continuous postsynaptic stimulation [13%]
- ☐ C. Electrical, because the charge of the current is the same in both the pre- and postsynaptic neurons [20%]
- ☐ D. Electrical, because electrical synapses transmit signals bidirectionally [5%]

Correct

60% answered correctly

Explanation:



Synapses in the nervous system can be either **electrical or chemical**. Information transmitted across electrical synapses is conveyed in the form of an ionic current that passes directly from the presynaptic neuron to the postsynaptic neuron via a **gap junction**, which is a cluster of channels that connects the cytoplasm of adjacent cells. Due to this cytoplasmic continuity and the mere 2-nm to 4-nm gap between the pre- and postsynaptic membranes, the current passes almost instantaneously from the presynaptic neuron to the postsynaptic neuron. Cytoplasmic continuity also allows signals (ions) to flow **bidirectionally** across the synapse.

In contrast, information transfer is typically **unidirectional** at chemical synapses, and the cytoplasm of the pre- and postsynaptic cells are not continuous. Depolarization of the presynaptic neuron results in the release of chemical messengers (**neurotransmitters**) that diffuse across the **synaptic cleft** (the 20-nm to 40-nm space between the pre- and postsynaptic membranes) and bind to postsynaptic membrane receptors to generate a new ionic current in the postsynaptic cell.

Signal transmission across chemical synapses is typically *slower than* transmission across electrical synapses because the signal must be modified (ie, electrical to chemical to electrical) and must cross a larger intercellular distance. Therefore, **Set A** depicts a **chemical synapse** because there is **a delay between the pre- and postsynaptic action potentials (APs)**. Set B, which shows nearly instantaneous firing of the postsynaptic cell, corresponds to an electrical synapse.

(Choice B) In response to a single presynaptic signal, continuous postsynaptic stimulation (represented by multiple postsynaptic APs) can occur at chemical synapses when **neurotransmitter clearance** from the synapse is impaired. However, this characteristic is *not* present in either graph set.

(Choice C) Changes in membrane potential can occur in the same direction for presynaptic and postsynaptic cells of both electrical and chemical synapses. However, action potentials transmitted through chemical synapses can also cause the postsynaptic cell to change membrane potential in a *different* direction than the presynaptic cell (ie, either depolarize or hyperpolarize, depending on the receptor activated). Both graph sets depict postsynaptic depolarization, so this feature cannot be used to distinguish the type of synapse.

(Choice D) The fact that Set A corresponds to a chemical synapse cannot be deduced based on the direction of signal transmission because both graph sets depict unidirectional AP transmission.

Educational objective:

Electrical synapses transfer information from one cell to another via passive ionic current flow through gap junctions. In contrast, chemical synapses use neurotransmitters to transfer information, which is a slower process.

Time spent: 0 sec
Subject:
 Biology

QID: 400951
System:
 Endocrine and Nervous Systems

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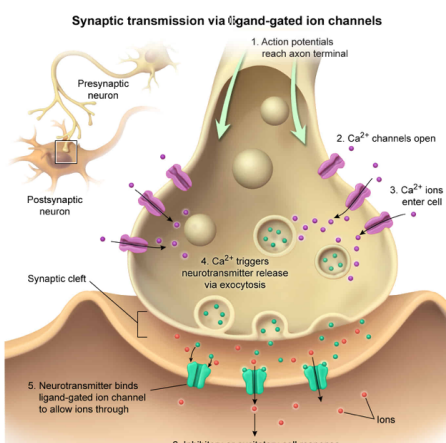
Which of the following synaptic events would likely contribute most directly to the generation of an action potential in a postsynaptic neuron?

- ☐ A. The presynaptic neuron releases large quantities of gamma-aminobutyric acid (GABA). [4%]
- ✓ ☒ B. Postsynaptic neurotransmitter-gated Na^+ and Ca^{2+} channels open. [70%]
- ☐ C. Neurotransmitter reuptake into the presynaptic cell and surrounding glia is increased. [5%]
- ☐ D. Neurotransmitter binding activates G protein–coupled receptors on the postsynaptic membrane. [19%]

Correct

70% answered correctly

Explanation:



In **chemical synapses**, when an action potential (AP) reaches the axon terminal of the presynaptic neuron, the terminal membrane depolarizes and causes **voltage-gated Ca^{2+} channels** to open. Extracellular Ca^{2+} ions flow down their electrochemical gradient into the axon terminal and stimulate neurotransmitter-containing **synaptic vesicles** to fuse with the presynaptic membrane and release their contents into the synaptic cleft by **exocytosis**. The neurotransmitters diffuse across the synaptic cleft and bind to receptors on the postsynaptic cell membrane, which can be ligand-gated ion channels or **G protein–coupled receptors** (GPCRs).

In most cases when a specific ligand (neurotransmitter) binds to its corresponding ligand-gated ion channel on the postsynaptic neuron, the channel changes its conformation to the open state. Depending on the type of channel and the environmental conditions (eg, membrane potential) in which the channel is found, ionic movement across the membrane results in either postsynaptic depolarization (excitation) or hyperpolarization (inhibition).

Because Ca^{2+} and Na^{+} concentrations are typically greater outside the neuron than inside, if ligand-gated ion channels permeable to these ions were to open in response to neurotransmitter release, both Ca^{2+} and Na^{+} would rush into the neuron and depolarize the membrane. This depolarization, if summed with other excitatory signals, would contribute *directly* to the generation of a postsynaptic AP.

(Choice A) Of the major **neurotransmitters**, gamma-aminobutyric acid (GABA) is the primary *inhibitory neurotransmitter* of the central nervous system. Increased release of GABA by the presynaptic neuron and its subsequent binding to postsynaptic receptors is likely to *prevent* the generation of an AP in the postsynaptic neuron.

(Choice C) *Increased* clearance of the neurotransmitter from the synapse (due to rapid reuptake or enzymatic degradation) would likely *decrease* the neurotransmitter's effect (eg, depolarization) on the postsynaptic cell. However, the effect of this neurotransmitter (ie, excitatory or inhibitory) is *not* specified.

(Choice D) Activation of GPCRs can promote depolarization or hyperpolarization, depending on the specific GPCR. In addition, GPCRs generally rely on signal transduction via **second messengers**, so their contribution to depolarization would likely be *less* direct than opening ligand-gated ion channels for more immediate ion flow. Furthermore, *neither* the neurotransmitter type *nor* the action of the postsynaptic GPCR (whether it causes depolarization or hyperpolarization) is specified; therefore, the effect of ligand binding to the GPCR *cannot* be determined.

Educational objective:

Although trends in neurotransmitter function exist, the postsynaptic response is ultimately determined by the type of receptor activated. Receptors can be either ligand-gated ion channels or G protein–coupled receptors.

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Subject:
Biology

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System:
Endocrine and Nervous Systems